

7.2 Software Support Processes

NOTE — The support processes listed under this subclause are specific to software and are labelled **Software Support Processes**. Although they play an integral role in assisting the Software Implementation Process, they may also provide services to other processes, such as **Agreement Processes, System Qualification Testing, Software Acceptance Support, Software Operation, and Software Maintenance**.

7.2.1 Software Documentation Management Process

The Software Documentation Management Process is a specialization of the **Information Management Process** from the Project Process Group in this International Standard. It governs the **planning, development, production, distribution, maintenance, and configuration management of all software-related documentation** produced throughout the software life cycle.

NOTE — ISO/IEC 15289 provides more detailed content for life cycle process information items (documentation).

7.2.1.1 Purpose

The purpose of the Software Documentation Management Process is to **develop and maintain recorded software information** produced by a process, ensuring that it is accurate, consistent, accessible, and traceable across the software life cycle.

7.2.1.2 Outcomes

As a result of successful implementation of the Software Documentation Management Process:

- a) a **documentation strategy** identifying the documentation to be produced during the life cycle of the software product or service is developed;
- b) the **standards and formats** to be applied for the development of software documentation are identified;
- c) required documentation for each process or project is identified;
- d) the **content and purpose** of all documentation is specified, reviewed, and

approved;

e) documentation is developed and made available in accordance with identified standards; and

f) documentation is maintained in accordance with defined criteria for retention, versioning, and change control.

7.2.1.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Documentation Management Process.

7.2.1.3.1 Process Implementation

This activity consists of the following task:

7.2.1.3.1.1

A **documentation plan**, identifying the documents to be produced during the life cycle of the software product, shall be developed, documented, and implemented. For each identified document, the following shall be addressed:

a) title or name;

b) purpose and content;

c) intended audience;

d) procedures and responsibilities for inputs, development, review, modification, approval, production, storage, distribution, maintenance, and configuration management;

e) schedule for intermediate and final versions.

7.2.1.3.2 Design and Development

This activity consists of the following tasks:

7.2.1.3.2.1

Each identified document shall be designed in accordance with applicable documentation standards covering **medium, content description, page numbering, figure/table placement, proprietary or security markings, packaging, accessibility, and other presentation requirements**.

7.2.1.3.2.2

Automated documentation generation tools may be used where appropriate. Input data used for document generation shall be **validated for correctness and completeness** before processing.

7.2.1.3.2.3

Prepared documents shall undergo **format, technical, and linguistic reviews** by qualified personnel prior to approval and release. Review outcomes shall be documented.

7.2.1.3.3 Production

This activity consists of the following tasks:

7.2.1.3.3.1

Documents shall be produced and distributed in accordance with the documentation plan. Production and distribution may use paper, electronic, or other media.

All documents shall include appropriate **metadata (e.g., version number, date, author, classification)**.

7.2.1.3.3.2

Master copies shall be stored according to organizational requirements for **record retention, security, maintenance, and backup**.

7.2.1.3.3.3

Controls shall be established in accordance with the **Software Configuration Management Process (subclause 7.2.2)** to ensure controlled baselining, versioning, and retrieval of documentation.

7.2.1.3.4 Maintenance

This activity consists of the following task:

7.2.1.3.4.1

When documentation is to be modified, the tasks of the **Software Maintenance Process** (subclause 6.4.10) shall be performed.

For documents under configuration management, modifications shall be:

- initiated through approved change requests,

- reviewed and re-approved according to the documentation plan, and
 - managed in accordance with the **Software Configuration Management Process (subclause 7.2.2)**.
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7.2.1.4 Deliverables

The following deliverables shall be produced during this process:

- Documentation Plan;
 - Document design templates and standards;
 - All identified software documentation (e.g., design docs, test plans, audit logs, release notes);
 - Review records and approval logs;
 - Metadata indexes or documentation registries;
 - Configuration-managed master copies;
 - Documentation Change Logs.
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7.2.1.5 Audit and Configuration Control

All documentation shall be placed under appropriate **configuration management**. Audit evidence (e.g., document versions, approval records, metadata logs) shall be retained in accordance with organizational policy or regulatory requirements. Digital signatures or equivalent mechanisms shall be applied to ensure integrity and authenticity. Documentation baselines shall be re-established after each approved change.